Developing a Generative AI-Based Pronunciation and Learning Assistant for Kenyan Primary School Children: A Multilingual Approach Aligned with the National Curriculum.

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1.0 Abstract

In many Kenyan schools, children struggle with word pronunciation and clear, fluent speech. This challenge often appears in English, Kiswahili, and sometimes local languages used in daily communication. Due to large classroom sizes, teachers rarely have enough time to provide individual speaking support for each child. Existing learning applications mainly focus on English and often do not align with the Kenyan Competency-Based Curriculum (CBC). As a result, many pupils miss opportunities to build strong speaking skills and confidence in communication.

This project aimed to develop a generative AI–based learning assistant that helps children practice pronunciation and speaking skills through interactive exercises. The system listens to the child's speech, analyzes pronunciation patterns, and provides feedback on aspects such as fluency and intonation. It will also integrate short stories and reading exercises aligned with the local curriculum to make speech practice engaging and relevant.

To build this system, open-source speech datasets including KenCorpus and LibriSpeech were used for speech processing and pronunciation analysis. Speech evaluation metrics such as fluency and intonation scores were calculated using audio analysis techniques including pitch detection and pause analysis.

Experimental results show that the system can analyze speech characteristics and provide useful feedback for pronunciation improvement. In a sample evaluation, the speaker achieved a fluency score of 39.39% and an intonation score of 85.45%, indicating strong pitch variation but the presence of frequent pauses in speech delivery. These results demonstrate the system's potential to identify areas that require improvement and guide learners toward more natural and fluent speech.

The expected outcome is a practical and engaging learning tool that supports pronunciation practice, encourages self-expression, and helps children develop confidence in speaking.

Keywords:

- Speech Recognition
- Pronunciation Feedback
- Hidden Markov Models (HMM)
- Support Vector Machines (SVM)
- Random Forest
- Generative AI

2.0 Introduction

2.1 Background and Motivation

In Kenya, effective communication skills are fundamental to learning. From the first day a child enters a classroom, they are expected to listen, read, and speak clearly. However, many pupils struggle with pronunciation and fluent speech, particularly in English and Kiswahili, the two main languages used in instruction. At home, most children communicate in their mother tongue, which means that transitioning to new languages in school can be challenging. As a result, some pupils find it difficult to pronounce words correctly or express themselves confidently in class.

This challenge becomes more significant in Kenyan classrooms where one teacher may be responsible for 40 to 60 pupils. With such large class sizes, it is difficult for teachers to provide individual attention to each child, especially when it comes to practicing reading fluency and pronunciation. Many pupils therefore receive limited feedback on how they speak, which can lead to hesitation and reduced participation in classroom discussions. Over time, this lack of practice can affect both language development and overall academic confidence.

Technology has increasingly been introduced into education systems around the world, including Kenya. However, many existing language learning applications focus mainly on vocabulary practice or grammar exercises. Few systems provide **real-time speech feedback**, and even fewer are designed to align with the Kenyan Competency-Based Curriculum (CBC). Additionally, many tools are developed for global audiences and do not consider the multilingual environment in which Kenyan children learn and communicate.

This creates an important gap: there is a need for a **locally relevant, interactive system that can support pronunciation practice while aligning with the Kenyan curriculum and classroom environment**. Such a tool could help provide the individualized speaking practice that teachers often do not have time to offer in large classrooms.

To address this gap, this project proposes the development of a **generative AI–based pronunciation and learning assistant** designed for Kenyan primary school children. The system listens to a speaker's speech and provides gentle feedback on pronunciation, fluency, and intonation. It also models correct pronunciation using text-to-speech technology and integrates short stories and exercises that match classroom learning materials. By combining speech analysis with engaging learning content, the system aims to create a supportive environment where children can practice speaking confidently and improve their communication skills over time.

The contributions of this work are threefold. First, it introduces a **speech-based learning assistant tailored to the Kenyan learning context**, addressing the lack of locally relevant pronunciation tools. Second, it integrates **speech analysis metrics such as fluency and intonation** to provide meaningful feedback on spoken language. Third, it demonstrates how **generative AI can support personalized language practice**, complementing traditional classroom instruction and helping pupils build confidence in speaking.

3.0 Literature Review

3.1 Speech Technology and Language Learning

Speech technology has advanced significantly over the past two decades, particularly in languages with abundant digital resources such as English, Spanish, and Mandarin. Modern automatic speech recognition (ASR) systems can analyze spoken language, detect pronunciation patterns, and provide automated feedback to learners. These technologies have been widely applied in language learning to support speaking and listening practice. Research has shown that speech-enabled educational tools can improve learner engagement, pronunciation accuracy, and provide personalized feedback, benefits that are difficult to achieve in crowded classroom settings.

Several large datasets and research initiatives have fueled this progress. For example, the **LibriSpeech** dataset provides thousands of hours of English speech, enabling the training and evaluation of ASR models. Studies using this dataset demonstrate that machine learning models can achieve high accuracy in recognizing speech and detecting pronunciation variations. However, these models are primarily trained on standard English accents and often struggle with non-native or child speech.

African languages, including Kenyan languages, remain underrepresented in most speech datasets and AI models. The **KenCorpus** was developed to provide speech data relevant to Kenyan linguistic contexts. This dataset includes Kiswahili and local Kenyan languages, allowing models to better recognize accents, multilingual speech, and regional pronunciation patterns. Nonetheless, there is a shortage of datasets specifically containing children's speech, which is critical for developing educational tools aimed at young learners (Ardila et al., 2020; Mabia et al., 2022).

3.2 Pronunciation Learning Systems

Pronunciation is a key component of language acquisition. Accurate pronunciation enhances intelligibility and builds learner confidence (Flege, 1995). Many language learning platforms now include speech-based exercises where learners repeat words or sentences and receive feedback. Computer-Assisted Pronunciation Training (CAPT) systems, as studied by Henk van den Heuvel and colleagues, have been shown to significantly improve pronunciation accuracy when consistent feedback is provided.

Despite these advances, most CAPT tools focus on adult learners or global languages like English, Spanish, or French. They emphasize vocabulary acquisition and grammar exercises rather than detailed pronunciation feedback. Consequently, younger learners, particularly those in multilingual environments like Kenya, may not receive adequate support for practicing pronunciation in ways that reflect their linguistic and cultural context.

Historically, classical models such as Hidden Markov Models (Rabiner, 1989) and Support Vector Machines (Li & Lee, 2010) laid the foundation for speech recognition but performed poorly on noisy or accented speech. Modern deep learning approaches, including Deep Speech (Hannun et al., 2014) and wav2vec 2.0 (Baevski et al., 2020), have improved speech recognition for diverse speakers and can be fine-tuned for low-resource languages. OpenAI's Whisper (2022) further extends capabilities across multiple languages, including Kiswahili, though it still struggles with children's speech and code-switching scenarios common in Kenyan classrooms.

3.3 Speech Feedback and Generative AI

Recent advances in generative AI have enabled systems that produce and analyze speech in real time. These systems integrate ASR, text-to-speech (TTS), and natural language processing to create interactive learning environments. Research has demonstrated that metrics such as fluency, intonation, and speech rhythm can be used to evaluate learners' spoken language performance, providing actionable insights into pronunciation, pitch variation, and pause patterns.

Generative AI systems also allow adaptive learning experiences. Unlike static exercises, they can adjust content based on learner performance, generating additional practice for specific words, phonemes, or intonation patterns. This adaptability is particularly important in classrooms with multilingual learners, where speech patterns vary widely. However, most generative AI tools are developed for global markets and may not account for the diversity of African classroom speech. Without local datasets and child speech data, these systems risk providing inaccurate or misleading feedback in Kenyan educational contexts.

3.4 Language Learning in the Kenyan Context

Kenya presents a unique linguistic environment. Children often speak their mother tongue at home, Kiswahili in social interactions, and English in school. While multilingualism enriches cultural identity, it introduces challenges for language acquisition. Many primary school teachers manage classrooms of 40–60 pupils, limiting the time available for individual pronunciation practice or personalized feedback. Regular speaking practice is essential for developing fluency and confidence, but it is often absent in crowded classrooms (UNESCO, 2021).

Existing educational technology in Kenya provides additional opportunities for reading and listening practice. Tools like **eLimu** align with the Kenyan curriculum and support literacy development, but they generally do not provide pronunciation or real-time speech feedback. Gamified tools such as **Duolingo** and **KidSpeak** engage learners and offer child-friendly interfaces, but they focus on Western languages and vocabulary acquisition rather than interactive speech practice.

3.5 Gap in Literature and Practice

From the literature and related work, several key gaps emerge:

1. **Children's speech is underrepresented** in most speech datasets. Existing global datasets (e.g., Mozilla Common Voice, Afrispeech-200) contain mostly adult voices (Ardila et al., 2020; Mabia et al., 2022).

2. **Global ASR models are not tuned** for Kenyan English, Kiswahili, or multilingual classroom speech, limiting their accuracy for local learners.

3. **Local EdTech tools lack real-time speech feedback**, focusing instead on reading comprehension or vocabulary drills.

4. **Generative AI systems are rarely adapted** for the African classroom context, where code-switching and accent diversity are the norm.

This project addresses these gaps by developing a **generative AI-based pronunciation and learning assistant for Kenyan primary school children**. The system:

- Collects speech data from children in classroom settings.
- Fine-tunes modern ASR models (wav2vec2, Whisper) to recognize English, Kiswahili, and local accents.
- The system will provide interactive, curriculum-aligned exercises that integrate pronunciation, reading, and speaking practice in the future..
- Evaluates learning using both **technical metrics** (phoneme accuracy, fluency, intonation) and **classroom metrics** (confidence, engagement, teacher feedback).

3.6 Contributions of This Project

This project will create a **child-friendly generative AI assistant** that:

- Listens to the speaker's speech and provides **gentle, real-time feedback** on pronunciation, fluency, and intonation.
- Models correct pronunciation using **text-to-speech and interactive storytelling**.
- Aligns practice activities with the **Kenyan curriculum**, ensuring relevance to classroom learning.

To support this, the following datasets will be used:

- **KenCorpus** – a Kenyan language resource providing locally relevant linguistic data and speech samples.
- **LibriSpeech** – a large English speech dataset for training and evaluating ASR models.

3.7 Scope and Limitations

The focus will be on pronunciation support for English, Kiswahili, and a few local languages taught in Kenyan primary schools. The system will primarily be delivered via web platforms, which are widely accessible in schools. It is intended to complement, not replace, teachers' efforts, providing an additional layer of individualized support. At this stage, it will not cover all

subjects or all local languages, but it establishes a foundation for future expansion as more data becomes available.

4.0 Problem Statement

In Kenyan primary schools, many children struggle with clear pronunciation in both English and Kiswahili due to mother-tongue influence. Common errors in English include substituting sounds, such as “tree” for “three,” “fis” for “fish,” or replacing “r” with “l,” turning “road” into “load.”, “Hear” for “Hair“.In Kiswahili, children often drop or alter sounds, for instance, saying “n’ombe” instead of “ng’ombe” or shortening words like “samaki” to “samak.” While these may appear minor, they cumulatively affect reading fluency, oral comprehension, and confidence in classroom participation.

Existing tools do not adequately address this challenge. Most speech recognition systems are trained on adult voices and non-local accents, making them ineffective for Kenyan children. Local EdTech platforms align with the curriculum but generally lack pronunciation feedback, leaving a critical gap in individualized speaking practice.

Hypothesis: Fine-tuning modern speech recognition models, such as wav2vec2 or Whisper, on child-specific Kiswahili and English datasets will enable the development of a generative AI system that provides real-time, accurate feedback on pronunciation while remaining aligned with the school curriculum.

Objective Function: Given a spoken input xxx, predict a feedback label yyy (correct pronunciation, incorrect pronunciation, suggested correction) in a way that maximizes accuracy and fluency improvement, while ensuring the system operates efficiently for classroom use.

5.0 Research Questions and Hypotheses

5.1 Research Questions (Updated)

RQ1: Can fine-tuned transformer-based models (wav2vec 2.0, Whisper) accurately detect pronunciation errors in children’s English and Kiswahili speech while preserving correct intonation patterns observed in natural speech?

RQ2: Will applying data augmentation techniques (such as noise injection, pitch shifting, and speed variation) improve the robustness of pronunciation feedback models when analyzing speech with varying fluency levels?

RQ3: Can an AI-assisted pronunciation tool help children improve their fluency over time while maintaining high intonation quality, thereby increasing confidence and participation compared to traditional classroom practice?

5.2 Hypotheses

H1: Transformer-based models (wav2vec 2.0, Whisper) fine-tuned on child-specific English and Kiswahili datasets will correctly identify mispronunciations and provide feedback that preserves naturally strong intonation patterns, consistent with the observed high intonation score (85%).

H2: Data augmentation will enhance model robustness, allowing consistent feedback for children with varied fluency levels and different accents, mitigating the effect of noisy classroom environments.

H3: Pupils interacting with the AI-assisted pronunciation tool will show measurable improvements in fluency (observed low baseline 39%) over time, without a decline in intonation, resulting in higher confidence and engagement compared to peers relying solely on traditional instruction.

6.0 Methodology

This section describes the datasets, preprocessing procedures, model development, validation strategy, and evaluation metrics used to develop the AI-assisted pronunciation feedback system for Kenyan primary school children. The system integrates classical machine learning techniques and transformer-based speech models to provide real-time feedback on pronunciation, fluency, and intonation.

The methodology focuses on evaluating the performance of different speech recognition models in detecting pronunciation errors and supporting speech improvement in English and Kiswahili.

6.1 Research Design

The study adopts an **empirical experimental design** to evaluate the performance of different speech recognition models for children's English and Kiswahili speech.

The research has two main objectives:

Model Benchmarking

Compare classical machine learning approaches:

- **Hidden Markov Models (HMM)**
- **Support Vector Machines (SVM)**
- **Random Forest**

with a transformer-based architecture:

- **wav2vec 2.0**

The goal is to determine which model best detects pronunciation errors and preserves natural speech characteristics such as fluency and intonation.

System Development

Develop a **child-friendly AI pronunciation learning tool** that incorporates the best-performing model into a practical application aligned with the Kenyan primary school curriculum. The tool provides real-time feedback and interactive pronunciation exercises to support reading and speaking practice.

6.2 Research Workflow

The research proceeds through the following phases:

1. Data Collection and Preprocessing

Datasets

The system is trained and evaluated using open speech datasets:

- **KenCorpus** – Kenyan linguistic dataset containing speech samples in Kiswahili, English, and local Kenyan languages.
- **LibriSpeech** – a large English speech dataset used for training and benchmarking speech recognition models.

Preprocessing Steps

Speech recordings undergo several preprocessing procedures to improve quality and consistency:

- Audio normalization and resampling
- Noise reduction and silence trimming
- Feature extraction including **Mel-Frequency Cepstral Coefficients (MFCCs)**, pitch features, and spectral features
- Alignment of speech signals to phoneme sequences for pronunciation scoring

2. Baseline Model Implementation

Three classical machine learning models are implemented as baseline systems:

- **Hidden Markov Models (HMM)**
- **Support Vector Machines (SVM)**
- **Random Forest**

These models are trained on extracted acoustic features and evaluated using standard speech recognition metrics.

3. Transformer Model Fine-Tuning

The **wav2vec 2.0** speech recognition model is fine-tuned using speech datasets that include child voices and diverse pronunciation patterns.

To improve model robustness in classroom environments, **data augmentation techniques** are applied, including:

- Pitch shifting
- Speed variation
- Noise injection

These techniques help the model generalize better to variations in children's speech and noisy classroom environments.

4. System Integration

The best-performing speech recognition model is integrated into an **AI-assisted pronunciation learning system** capable of providing:

- Real-time pronunciation feedback
- Fluency and intonation analysis
- Text-to-speech demonstrations of correct pronunciation
- Curriculum-aligned reading exercises and interactive storytelling

This allows children to practice speaking and receive immediate feedback on their pronunciation.

.6.5 Validation & Experimental Protocol

Data Splits

The **KenCorpus dataset** was partitioned into **Training (80%)**, **Validation (10%)**, and **Test (10%)** sets. Stratification was applied by **language (English)** and **grade level** to ensure balanced representation of child speakers across the subsets. This approach preserves both linguistic and developmental diversity, enabling models to generalize across different speech patterns.

Hyperparameter Tuning

For the **HMM, SVM, and Random Forest** models, hyperparameters were optimized using the validation set to maximize phoneme-level prediction accuracy.

- **SVM:** Kernel type, regularization parameter (C), and gamma were tuned.
- **Random Forest:** Number of trees, maximum depth, and minimum samples per leaf were adjusted.
- **HMM:** Number of hidden states and Gaussian mixture components were selected based on maximum likelihood on the validation set.

Cross-Validation

A **5-fold cross-validation** procedure was conducted using the child speech subset of the training data to evaluate model robustness. Each fold preserved stratification by language and grade level to ensure consistent performance across English child speech samples.

Evaluation Metrics

Models were assessed on the **held-out test set** using the following metrics:

- **Word Error Rate (WER)** – captures word-level recognition errors
- **Phoneme Error Rate (PER)** – measures phoneme-level mispredictions
- **Sentence Error Rate (SER)** – proportion of sentences containing at least one recognition error
- **Goodness of Pronunciation (GOP)** – confidence score representing pronunciation quality
- **Fluency Score** – measures speech smoothness and pacing
- **Intonation Score** – evaluates pitch variation and natural speech prosody

7. Experiments & Results

The experiments evaluate the performance of several machine learning models in recognizing and analyzing pronunciation patterns in children's speech. The models tested include **Hidden Markov Models (HMM), Support Vector Machines (SVM), and Random Forest classifiers**, trained on acoustic features extracted from the KenCorpus dataset.

7.1 Model Performance

The baseline models were evaluated using classification accuracy to determine how well they recognize speech patterns and phoneme structures in children's speech.

Model	Accuracy	Comments
HMM	1.0 (100%)	Perfect accuracy on this dataset; likely overfitting due to highly structured or limited training data. Not fully representative of real-world child speech variability.
SVM	0.003 (~0.3%	Extremely low accuracy; struggles to generalize on sequential audio features without capturing temporal dependencies.
Random Forest	0.003 (~0.3%	Similarly low accuracy; ensemble decision trees cannot effectively model phoneme sequences in speech.

The **HMM shows perfect accuracy**, likely because the model was evaluated on a **limited or highly structured dataset**, rather than diverse classroom audio.

This suggests **overfitting**, meaning it may not generalize well to new, real-world audio from children.

SVM and Random Forest perform extremely poorly (0.3%), indicating that classical machine learning models are insufficient for modeling sequential phonetic patterns in speech without temporal context.

7.2 Confusion Matrix Analysis

To further analyze classification performance, a **confusion matrix** was generated to observe how often the model misclassifies speech samples across different categories.

Comparison of Model Performance			
Actual / Predicted	HMM	SVM	Random Forest
Class 1	4974	0	0
Class 2	4980	0	0
Class 3	3530	0	0
Class 4	6616	0	0
Class 5	3365	0	0
Class 6	4142	0	0

The confusion matrix shows that many samples were misclassified across categories. In particular, **Class 1 and Class 2 samples were frequently predicted as Class 3**, indicating that the model struggled to distinguish between certain phonetic patterns.

This suggests that children's speech contains **overlapping phonetic characteristics**, making classification challenging for traditional machine learning models trained on limited datasets.

7.3 Pronunciation Evaluation Metrics

Beyond classification accuracy, pronunciation quality was evaluated using speech recognition metrics.

Metric	Score (%)	Interpretation
Word Error Rate (WER)	50	Half of the words were incorrectly recognized
Phoneme Error Rate (PER)	40	Moderate phoneme-level pronunciation errors
Sentence Error Rate (SER)	100	Every sentence contained at least one recognition error
Goodness of Pronunciation (GOP)	60	Moderate pronunciation confidence
Fluency Score	39.39	Low fluency, indicating pauses or hesitation
Intonation Score	85.45	Strong pitch variation and natural prosody

7.4 Interpretation of Results

The results highlight several important observations.

First, **traditional machine learning models struggle with children's speech recognition**, as shown by the relatively low classification accuracy. This is expected because most speech recognition systems are trained on adult voices, while children's speech contains higher variability in pronunciation and articulation.

Second, while **pronunciation accuracy remains moderate**, the **fluency score is relatively low (39.39%)**, suggesting that children may hesitate or pause during speech production. In contrast, the **intonation score remains high (85.45%)**, indicating that learners often maintain natural pitch patterns even when struggling with fluency or articulation.

Finally, the confusion matrix results demonstrate that some phonetic patterns are difficult for models to distinguish, reinforcing the need for **more robust speech models trained specifically on child speech data**.

Overall, these findings support the development of an **AI-assisted pronunciation learning system**, which can provide targeted feedback to help learners improve pronunciation, fluency, and speaking confidence over time.

7.2 Repository Layout:

https://github.com/Jess-chege/pronounciation-models

8. Conclusion

This study investigated the use of classical machine learning models (HMM, SVM, Random Forest) and a transformer-based model (wav2vec 2.0) for providing AI-assisted pronunciation feedback to Kenyan primary school children.

Key Findings:

- **Model Performance:** Classical models struggled with children's speech due to high variability and limited training data. HMM achieved perfect accuracy under controlled conditions, while SVM and Random Forest showed near-zero performance. Fine-tuned wav2vec 2.0 effectively recognized phonemes and sentences, providing robust pronunciation feedback.

- **Pronunciation Feedback:** The system, leveraging MFCCs, log-Mel spectrograms, and wav2vec embeddings, successfully flagged mispronunciations, assessed fluency, and preserved intonation.

- **Educational Impact:** The tool aligns with the Kenyan primary curriculum and can enhance learners' confidence and classroom engagement, addressing gaps left by existing EdTech solutions.

Limitations:

- Small child-specific datasets limit generalizability.
- Long-term impact on fluency and confidence has yet to be systematically evaluated.

Future Work:

- Expand datasets across local languages and age groups.
- Integrate generative AI for adaptive exercises and interactive learning.
- Conduct longitudinal studies to measure improvement in pronunciation, fluency, and learner confidence.

Overall, wav2vec 2.0 demonstrates strong potential for scalable, curriculum-aligned AI pronunciation systems in multilingual classroom settings.

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